

Chapter 15

Applications Of Inversive Geometry

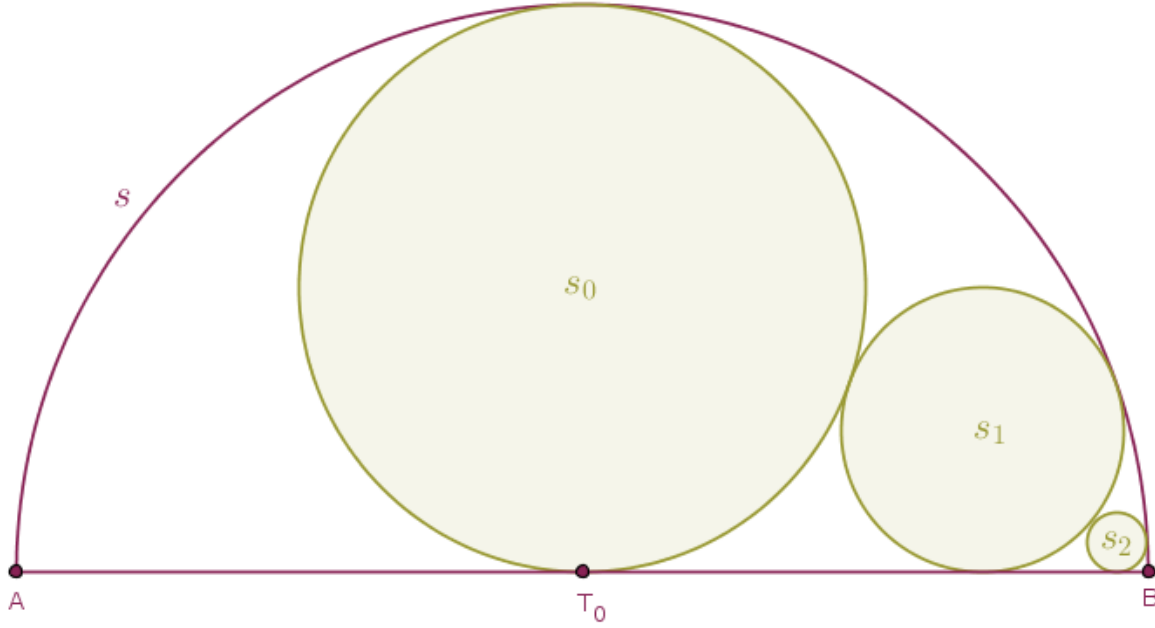
Discovery And Solution Of A Recurrence Relation

In this chapter we will see that in certain cases inversive geometry can not only generate but can also solve a homogeneous linear recurrence relation with constant coefficients by purely geometric means.

Put slightly differently, in this chapter we will see that the vaguely defined layers of mathematics that seem very different at first blush may possess the connections between that are hidden from plain sight and that are not at all obvious but are there nonetheless and all along.

Problem 21. Let let s be a semicircle subtending the diameter, AB , constructed on two distinct points, A and B , and let $s_0, s_1, s_2, \dots$ be the coplanar circles inscribed into the semicircle s in such a way that all the inscribed circles s_n touch the said diameter, AB , and the said semicircle, s , internally.

Moreover, let the, logically first, inscribed circle, s_0 , pass through the middle point, T_0 , of the given diameter AB and let the n -th inscribed circle s_n touch the, logically previous, circle, s_{n-1} , externally for any $n = 1, 2, 3, \dots$ as shown below (Figure 15.1):



Designate the lengths of the radii of the given circles $s, s_0, s_1, s_2, \dots$ with:

$$r, r_0, r_1, r_2, r_3, \dots$$

correspondingly.

Designate the ratios of the known length of the radius r of the semicircle s to the unknown lengths r_n of the radii of the inscribed circles s_n with t_n as follows:

$$t_0 = \frac{r}{r_0}, \quad t_1 = \frac{r}{r_1}, \quad t_2 = \frac{r}{r_2}, \quad t_3 = \frac{r}{r_3}, \dots$$

so that for any $n = 0, 1, 2, 3, \dots$ it is the case that:

$$t_n = \frac{r}{r_n}$$

Objectives:

a) recover the length r_n as a function of r and n

b) Prove that it is the case that: $t_0 = 2, t_1 = 4, t_2 = 18, t_3 = 100, t_4 = 578, t_5 = 3364$ and so on. In other words, show that for any positive whole number $n > 1$ the following recurrence relation holds:

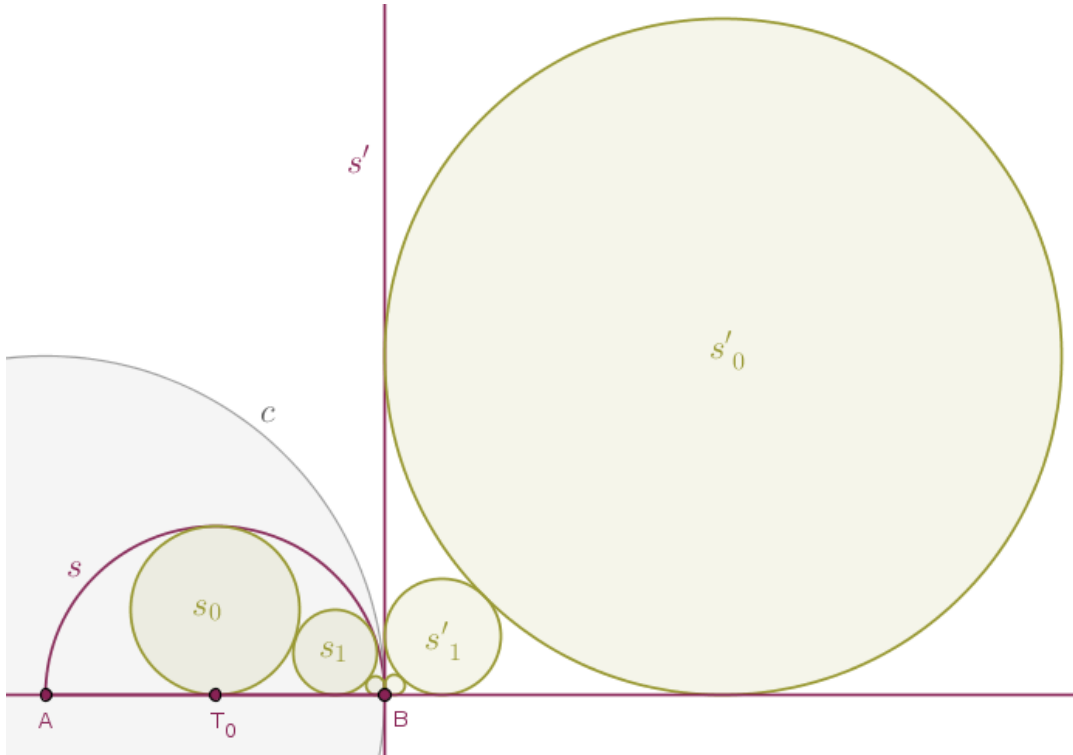
$$t_n = 6t_{n-1} - t_{n-2} - 4 \quad (1)$$

which implies that, astonishingly, all such numbers t_n are positive **integers**!

Solution. In order to address the first objective, **a)**, we try some inversions of the parent plane with respect to a circle; we, naturally, fail several times, but, eventually, on something like the try number three or four, we zoom in on a winner. Our plan of attack of this particular problem is to analyze the inversion of the given plane with respect to the circle, c , centered at the point A with the radius $|AB|$ with positive power, $c(A, |AB| = 2r)$.

Under the said inversion:

- the straight line AB will be transformed into itself (Theorem 9)
- the complete circle s will be transformed into a straight line, s' , (Theorem 17) that will be:
 - tangent to the circle of inversion, c , at the point B
 - orthogonal to the straight line AB
- the family of the inscribed circles s_n will be taken into the family of the inverted circles s'_n (Figure 15.2):



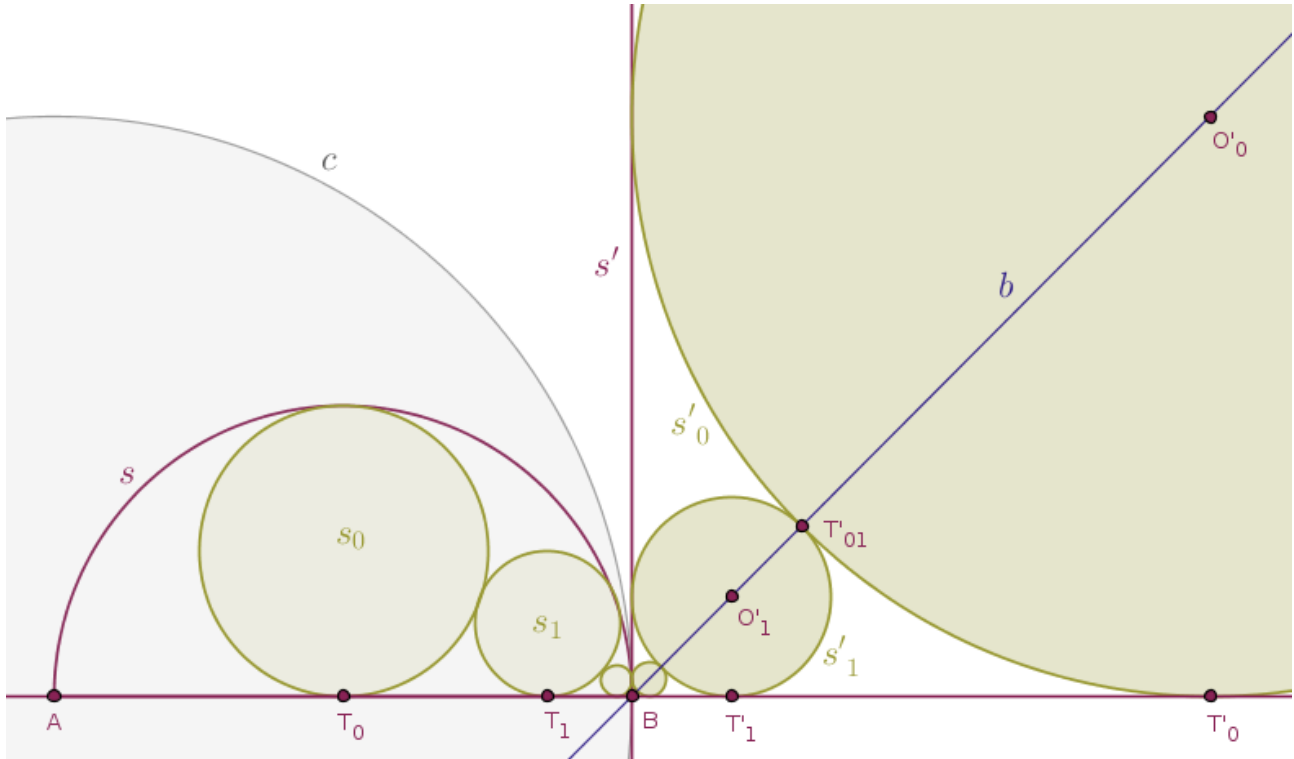
that will be inscribed into the right angle formed by the straight lines AB extended and s' , because under any inversion of the real plane with respect to a circle, the tangency points of the cooperating curves are also preserved (Theorems 37/38).

As such, after some number of failures, we managed to zoom in on an inversion that we referred to as a *winner*, because now we do have a tactical advantage. The problem of inscribing and analyzing the various features of the circles that touch the sides of a right angle is much simpler than the original problem.

For example, we know that the centers of all the inverted circles s'_n must be located on the angular bisector, b (shown below), of the right angle at the vertex B and that, computationally, the square root of two, $\sqrt{2}$, will simply have to pop up into the woodwork at least once somewhere.

Indeed. In this particular problem the image circle s'_0 can be drawn trivially as follows. Reflect the point T_0 in the circle of inversion, c , with positive power into the point T'_0 . We know how to do that with the Euclidean tools alone, see Problem 1.

That image point, T'_0 , will be located on the straight line AB . Erect a perpendicular to AB through T'_0 until that perpendicular intersects the angular bisector b at the point O'_0 . Draw the (green) circle s'_0 with the center at the point O'_0 and the radius $|O'_0T'_0|$ (Figure 15.3):



Repeat that maneuver for all other circles and recall that constructing a bisector of a given angle is a problem that can be solved with Euclidean tools alone as well, in 4 steps to be exact.

Now let $T_0, T_1, T_2, \dots$ be the points of tangency of the original circles $s_0, s_1, s_2, \dots$ with the diameter AB correspondingly and let $O'_0, O'_1, O'_2, \dots$ be the centers and $r'_0, r'_1, r'_2, \dots$ be the lengths of the radii of the image circles $s'_0, s'_1, s'_2, \dots$ correspondingly.

We remember that the tangency points of the cooperating curves under any inversion of the real plane with respect to a circle are preserved. Therefore, since the original circles s_n and the diameter AB touch at the above points T_n , the inverted circles s'_n will also touch the straight line AB extended at the corresponding points T'_n .

In other words, the above point T_0 will be taken by our inversion into the point T'_0 , the above point T_1 will be taken into the point T'_1 and so on. This was the idea for constructing the family of the image circles s'_n that we used above.

Thus, for the pair of the inversive points T_0 and T'_0 , according to Condition 3 of Definition 1 and our definition of a positive power of inversion, we have:

$$|AT_0| \cdot |AT'_0| = 4r^2 \quad (2)$$

from where:

$$|AT'_0| = \frac{4r^2}{|AT_0|} = \frac{4r^2}{r} = 4r$$

Clearly, for the length of the radius r'_0 of the inverted circle s'_0 we have:

$$\begin{aligned} r'_0 &= |BT'_0| = |AT'_0| - |AB| = 4r - 2r = 2r \\ r'_0 &= 2r \end{aligned} \quad (3)$$

Next, we observe that in this case **all** the inverted circles s'_n are homothetically similar under the homotheties of the parent plane centered at the point B . Thus, we would like to express the multitude of the coefficients of these homotheties via one computationally convenient magnitude.

One such convenient magnitude is the ratio of the lengths of the radii of the first two inverted circles $s'_{0,1}$, which are homothetically similar under the corresponding homothety of the parent plane centered at the point B , which implies that it is the case that:

$$\frac{r'_1}{r'_0} = \rho = \text{const} \in \mathbb{R} \quad (4)$$

But the length of the radius r'_0 is already known to us, see (3). Hence, it is convenient to express the lengths of all the consequent inverted radii in terms of r'_0 :

$$r'_1 = r'_0 \cdot \rho \quad (5)$$

Now we reason in terms of pairs of circles. The next two circles $s'_{2,3}$ are homothetically similar to the first pair of circles $s'_{0,1}$ under the same homothety:

$$\frac{r'_2}{r'_1} = \frac{r'_1}{r'_0} = \rho$$

from where via (5):

$$r'_2 = r'_1 \cdot \rho = r'_0 \cdot \rho \cdot \rho = r'_0 \cdot \rho^2 \quad (6)$$

Likewise, we claim that for the length of the radius r'_3 we will have:

$$r'_3 = r'_0 \rho^3$$

and so on, for any $n > 1$ we shall have:

$$r'_n = r'_0 \rho^n$$

In order to unmask the magnitude ρ , we observe that, clearly, on the one hand:

$$|BO'_0| = r'_0 \sqrt{2}$$

and on the other hand:

$$|BO'_0| = |BO'_1| + |O'_1O'_0|$$

But:

$$|BO'_1| = r'_1 \sqrt{2}$$

and:

$$|O'_1O'_0| = r'_1 + r'_0$$

Hence, exercising the concept of double-counting and equating the two different expressions for one and the same magnitude, we find:

$$r'_0 \sqrt{2} = r'_1 \sqrt{2} + r'_1 + r'_0$$

Divide the last relation through by r'_0 and use (4):

$$\sqrt{2} = \rho\sqrt{2} + \rho + 1$$

Solve the last equation for ρ :

$$\rho = \frac{\sqrt{2}-1}{\sqrt{2}+1} = 3 - 2\sqrt{2} \quad (7)$$

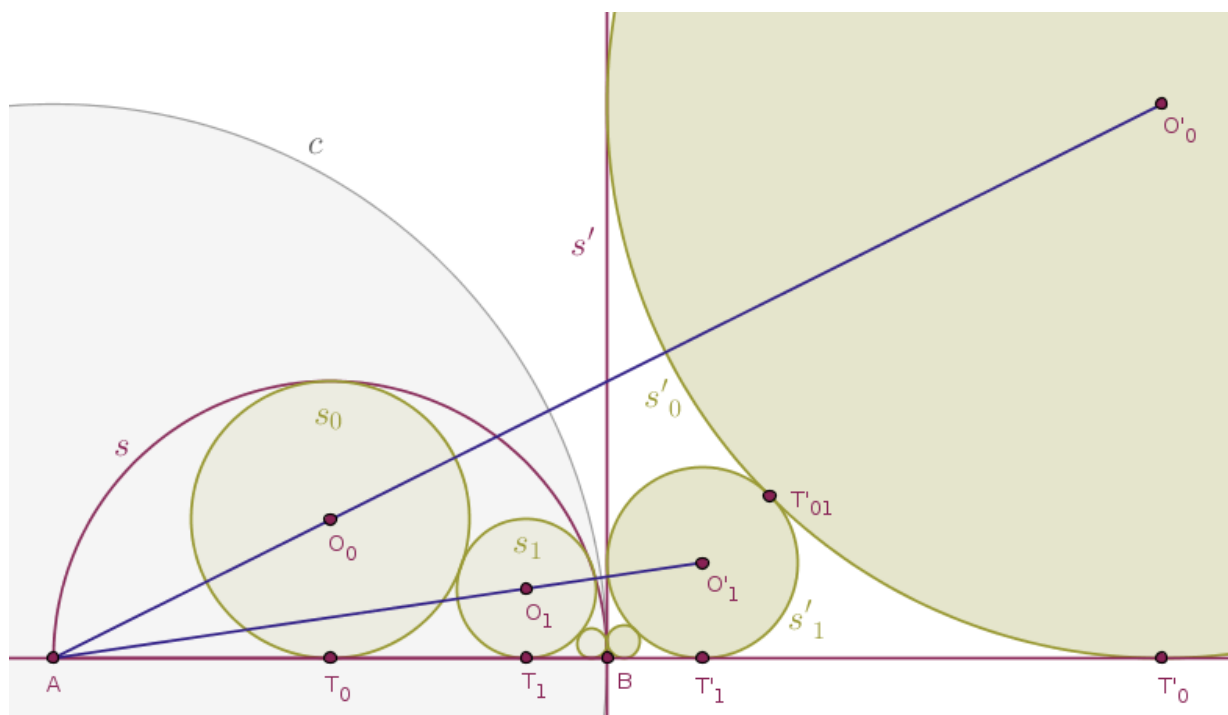
via multiplication of both the numerator and the denominator of the above ratio by the conjugate image of its denominator, $\sqrt{2} - 1$. Thus, for any $n = 0, 1, 2, 3, \dots$ it is the case that:

$$r'_n = 2r \cdot \left(\frac{\sqrt{2}-1}{\sqrt{2}+1} \right)^n \quad (8)$$

or, clearing the above denominator of the radicals:

$$\boxed{r'_n = 2r \cdot \left(3 - 2\sqrt{2}\right)^n} \quad (9)$$

Lastly, in order to express the length of r_n in terms of the length r'_n , we observe that pairwise the circles s'_n and s_n are also homothetically similar with respect to the center A (Figure 15.4):



That is, if we uniformly contract or shrink the parent plane with respect to the center A by the coefficient k such that:

$$k = \frac{|AT_n|}{|AT'_n|}$$

then the circle s'_n will be taken into the circle s_n . Since the points T_n and T'_n are the inverses of each other, it follows that:

$$|AT_n| \cdot |AT'_n| = 4r^2$$

implying that:

$$k = \frac{4r^2}{|AT'_n|^2} \quad (10)$$

Due to the discussed earlier homothetical similarity of the circles s'_n :

$$\frac{|BT'_n|}{|BT'_0|} = \frac{r'_n}{r'_0} = \rho^n$$

from where:

$$|BT'_n| = |BT'_0| \cdot \rho^n = 2r\rho^n$$

For the distances $|AT'_n|$ we, clearly have:

$$|AT'_n| = |AB| + |BT'_n| = 2r + 2r\rho^n = 2r(1 + \rho^n)$$

Thus, for k in (10), we find:

$$k = \frac{4r^2}{4r^2(1 + \rho^n)^2} = \frac{1}{(1 + \rho^n)^2}$$

and, finally, for r_n in terms of r'_n we have:

$$\begin{aligned} r_n &= k \cdot r'_n = \\ &= 2r \cdot \frac{\rho^n}{(1 + \rho^n)^2} \end{aligned}$$

which, after some lightweight manipulations of opening up the square, yields:

$$r_n = 2r \cdot \frac{\rho^n}{1 + 2\rho^n + \rho^{2n}}$$

and dividing both sides of the fraction by ρ^n , becomes:

$$\boxed{r_n(r, n) = \frac{2r}{\rho^{-n} + 2 + \rho^n}} \quad (11)$$

which concludes the solution of the objective **a**) since ρ is just a real number given by (7). If our hearts so desire, we could rewrite (11) more compactly:

$$r_n(r, n) = 2K_n r, \quad K_n = \frac{1}{\rho^{-n} + 2 + \rho^n}$$

and so on.

Speaking about hidden connections between seemingly disparate layers of mathematics. It turns out that in integral calculus we have the following result:

$$\int_0^{\pi/2} \frac{x \, dx}{\sin(x) + \cos(x)} = \frac{\pi\sqrt{2}}{8} \cdot \log\left(\frac{\sqrt{2} + 1}{\sqrt{2} - 1}\right)$$

where our ratio ρ from (7) makes a prominent appearance.

Onto the proof of the objective **b**). Due to (11), for the ratios t_n we have for any $n = 0, 1, 2, 3, \dots$:

$$2t_n = \frac{2r}{r_n} = \rho^{-n} + 2 + \rho^n \quad (12)$$

which, with ρ taken from (7), yields:

$$2t_n = \left(3 + 2\sqrt{2}\right)^n + 2 + \left(3 - 2\sqrt{2}\right)^n \quad (13)$$

because, clearly, the magnitudes of ρ and ρ^{-1} are the conjugate images of each other:

$$\frac{1}{3 - 2\sqrt{2}} = 3 + 2\sqrt{2}$$

which, further, implies that the sum of these conjugate images comes out to be just a positive integer:

$$\rho + \rho^{-1} = 3 - 2\sqrt{2} + 3 + 2\sqrt{2} = 6$$

and that the product of these conjugate images is a positive integer also:

$$(3 - 2\sqrt{2}) \cdot (3 + 2\sqrt{2}) = 1$$

Working the Vieta's formulas in reverse order, we assemble the corresponding quadratic equation of which our conjugates are the roots:

$$x^2 - 6x + 1 = 0 \tag{14}$$

from where we also conclude that it must be the case that:

$$\rho^2 - 6\rho + 1 = 0 = \frac{1}{\rho^2} - \frac{6}{\rho} + 1 \tag{15}$$

by the definition of a root of an equation.

We note in passing that the equation shown in (14) happens to coincide with the so-called **characteristic equation** of the following homogeneous linear recurrence relation with constant coefficients of order two: $t_n = 6t_{n-1} - t_{n-2}$.

In order to prove that for any whole positive number $n > 1$ the following recurrence relation holds:

$$t_n = 6t_{n-1} - t_{n-2} - 4$$

we manipulate its doubled and a slightly rearranged image as per our finds in (12, 15) as follows:

$$\begin{aligned} 2t_n - 6 \cdot 2t_{n-1} + 2t_{n-2} + 8 &= \\ &= \left(\frac{1}{\rho^n} + 2 + \rho^n \right) - \\ &- 6 \left(\frac{1}{\rho^{n-1}} + 2 + \rho^{n-1} \right) + \\ &+ \left(\frac{1}{\rho^{n-2}} + 2 + \rho^{n-2} \right) + 8 \end{aligned}$$

Grouping the direct powers of ρ together, grouping the reciprocal powers of ρ together and noting that $2 - 12 + 2 + 8 = 0$, we see that:

$$2t_n - 6 \cdot 2t_{n-1} + 2t_{n-2} + 8 =$$

$$= \rho^{n-2} (\rho^2 - 6\rho + 1) + \frac{1}{\rho^{n-2}} \left(\frac{1}{\rho^2} - \frac{6}{\rho} + 1 \right) = 0$$

due to (15). Therefore, it is indeed the case that for any whole positive number $n > 1$ the following recurrence relation holds:

$$t_n = 6t_{n-1} - t_{n-2} - 4$$

which is what was required to prove.

We can now carry out the finite computations for t_n to our heart's content, as per (1):

$$2t_0 = \left(3 + 2\sqrt{2}\right)^0 + 2 + \left(3 - 2\sqrt{2}\right)^0 = 4$$

$$2t_1 = \left(3 + 2\sqrt{2}\right)^1 + 2 + \left(3 - 2\sqrt{2}\right)^1 = 8$$

$$2t_2 = \left(3 + 2\sqrt{2}\right)^2 + 2 + \left(3 - 2\sqrt{2}\right)^2 = 36$$

$$2t_3 = \left(3 + 2\sqrt{2}\right)^3 + 2 + \left(3 - 2\sqrt{2}\right)^3 = 200$$

$$2t_4 = \left(3 + 2\sqrt{2}\right)^4 + 2 + \left(3 - 2\sqrt{2}\right)^4 = 1156$$

Thus, officially, with the following initial conditions:

$$t_0 = 2, \quad t_1 = 4$$

the following function $t(n)$:

$$t(n) = 1 + \frac{\left(3 + 2\sqrt{2}\right)^n + \left(3 - 2\sqrt{2}\right)^n}{2}$$

is the solution of the following non-homogeneous linear recurrence relation with constant coefficients:

$$t_n = 6t_{n-1} - t_{n-2} - 4$$

for all whole positive numbers $n > 1$. $\square$